2-HOUR HANDS-ON CLASS · ETL WITH EUROSTAT DATA

ETL: from raw data to useful data

Extract · Transform · Load

THE PROBLEM

Real data does not arrive clean

geo	2000	2001	2002
BE	: u	5.99 i	6.09 d
EL	3.71 i	3.50 i	3.57 i
EU28	:	:	5.00 e

Real excerpt of the Eurostat file (public spending on education, % of GDP)

: means "not available", not zero.

i, d, e are notes on data quality, glued to the number.

EU28 is not a country, it is an aggregate.

THE SOLUTION

ETL has three phases

E

Extract

Get the data from the source.

Buy the ingredients

T

Transform

Clean and reshape the data.

Cook

L

Load

Leave the result where it can be used.

Serve the dish

PHASE 1 · EXTRACT

Data can come from many places

- CSV and TSV files
- JSON and XML
- Excel sheets
- Databases
- Web APIs
- Web pages

Today: a TSV file from Eurostat, which you will read into a table with Python.

PHASE 2 · TRANSFORM

This is where the decisions are

Convert values
From text to number, split off the flag.

Handle missing values
Leave, drop or fill them.

Select
Keep the countries and years of interest.

Join tables
Add the country name from its code.

Aggregate
Means per country, sums per year.

Sort and rank
Who is first and who is last.

Every transformation is a decision, and it changes the result.

PHASE 3 · LOAD

The result must go somewhere useful

CSV
Universal and readable by everyone.

Parquet
Compact and keeps the types.

Database
Can be queried and shared.

Charts
To communicate the result.

If someone else can repeat the process, the result can be trusted.

AN EXAMPLE

One cell, five columns

BEFORE (RAW)

`5.99 i`

A number and a note, inside a single text.



AFTER (CLEAN)

geo	country	year	value	flag
BE	Belgium	2001	5.99	i

Each piece of data has its own column, and the number is now a number.

THE DATA STRUCTURE

A DataFrame is a table with types and labels

country	year	value
Denmark	2010	8.81
Spain	2010	4.98
Finland	2010	6.85

Public spending on education, % of GDP

Columns: each has a single type (text, integer, decimal).

Rows: each is one observation.

Labels: rows and columns have names.

Like a matrix with names, or a strict spreadsheet.

HOW WE WILL WORK

Read, run, try

Read
Each step is explained before its code.



Run
Shift + Enter, in order, from the top.



Try
Fill the gaps in "Your turn" cells and run the check.

Got a red error? Read the last line.

SESSION PLAN

Two hours, six parts

Time	Part	What we do
0–15 min	Slides and setup	ETL concept, open the notebook
15–30 min	1 · DataFrames	pandas in ten minutes
30–50 min	2 · Extract	Look at the files, read CSV and TSV
50–90 min	3 · Transform	Clean, decide, check, find the bugs
90–110 min	4 · Load	Save and chart
110–120 min	5 · Reflection	Review and wrap up

BEFORE OPENING ANYTHING

Write down your prediction

1. How many values do you think are missing from the file?
2. Which European country spends the most on education?
3. What do you think the original file looks like?

Write it on a sheet of paper. We will check it at the end.